

Q.1) Constructing Gaussian Beams from plane waves**(150 marks)**

Go through the shared MATLAB code for constructing a TE polarized Gaussian beam in a uniform medium by taking a suitable superposition of plane waves at frequency ω . Note that I have taken $k_z \gg \sqrt{k_x^2 + k_y^2}$.

Modify the code to numerically compute and plot the Poynting vector to show that the Gaussian beam propagates (approximately) along the z direction.

Now write a similar code for the case of inhomogeneous medium with a planar interface between air and glass ($n = 1.5$). Your code should be able to compute and plot the total electric field distribution at any given plane parallel to the interface. Make it general to handle oblique incidence.

Q.2) TE modes of a dielectric slab waveguide**(50 marks)**

Revisit the derivation of modes propagating at an angle greater than the critical angle in a dielectric slab of finite thickness (surrounded by a lower index medium on both sides). Complete all the missing steps in the derivation, i.e., derive the relations between the coefficients leading to even or odd nature of the field distribution with respect to the z-coordinate. Also state the even/odd nature of all the components of electric and magnetic field.

Analytically determine the time-averaged Poynting vector for the TE even mode both inside and outside the dielectric slab. Comment on its components parallel and perpendicular to the interface.